Python While Loop

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While Loop is used to execute a block of statements repeatedly until a given condition is satisfied. When the condition becomes false, the line immediately after the loop in the program is executed.

Here, the condition for while will be True as long as the counter variable (count) is less than 3.

```
count = 0
while count < 3:
    count = count + 1
    print("Hello Geek")
```

```
Hello Geek
Hello Geek
Hello Geek
```

Output

```
Hello Geek
Hello Geek
Hello Geek
```

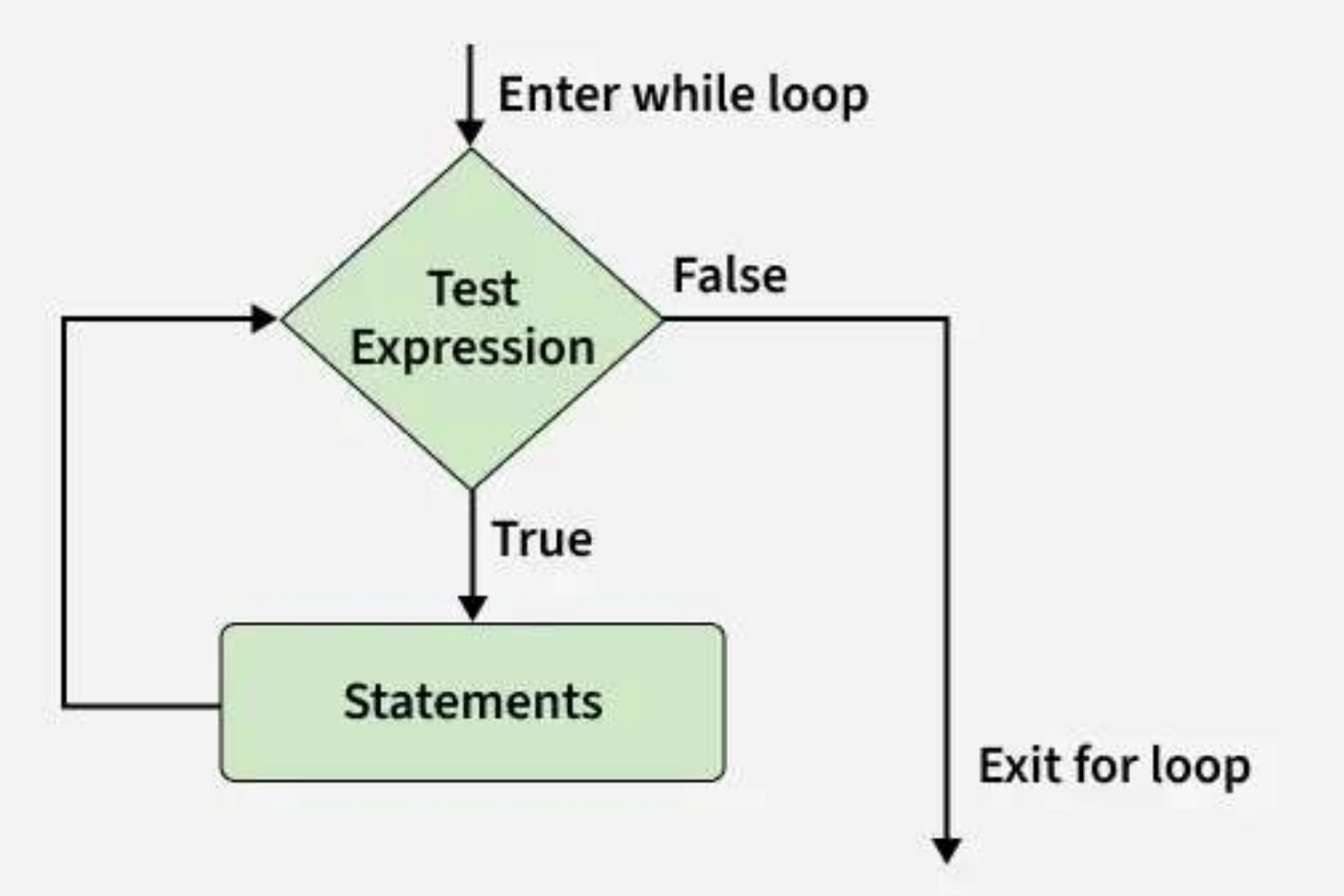
Syntax

```
while expression:
    statement(s)
```

Parameters:

- **condition** a boolean expression. If it evaluates to True, code inside the loop will execute.
- **statement(s)** that will be executed during each iteration of the loop.

Flowchart of While Loop



While Loop

Infinite while Loop

An infinite loop is a loop that keeps running continuously because its condition always remains True. Such loops do not stop on their own and continue executing until the program is manually terminated.

```
age = 28
while age > 19:
    print('Infinite Loop')
```

Explanation: Here, the condition age > 19 is always True because the value of age never changes inside the loop. Therefore, the loop runs infinitely.

Using with continue statement

Continue statement is used to skip the current iteration of the loop and move directly to the next iteration.

```
i = 0
a = 'geeksforgeeks'

while i < len(a):
    if a[i] == 'e' or a[i] == 's':
        i += 1
        continue
    print(a[i])
    i += 1
```

```
g
k
```

```
f
o
r
g
k
```

Output

```
g
k
f
o
r
g
k
```

Explanation: Here, whenever the character is 'e' or 's', the loop skips printing it and continues with the next character.

Using with break statement

Break statement is used to immediately terminate the loop when a specific condition becomes True.

```
i = 0
a = 'geeksforgeeks'

while i < len(a):
    if a[i] == 'e' or a[i] == 's':
        i += 1
        break
    print(a[i])
    i += 1
```

```
g
```

Output

```
g
```

Explanation: Here, the loop stops as soon as it encounters the character 'e' or 's'.

Using with pass statement

pass statement is a null statement. It does nothing when executed and is mainly used as a placeholder for future code.

```
a = 'geeksforgeeks'
i = 0
while i < len(a):
    i += 1
    pass
```

```
print('Value of i :', i)
```

Output

```
Value of i : 13
```

Explanation: Here, the loop runs through all characters of the string, but the pass statement performs no action inside the loop body.

Using with else

else block in a while loop executes only when the loop finishes normally without encountering a break statement. In first example, loop completes all iterations, so the else block executes. In second example, loop stops because of break, so the else block is skipped.

```
i = 0
while i < 4:
    i += 1
    print(i)
else:
    print("No Break\n")

i = 0
while i < 4:
    i += 1
    print(i)
    break
else:
    print("No Break")
```

```
1
2
3
4
No Break

1
```

Output

```
1
2
3
4
No Break

1
```

While Loops in Python

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Suggested Quiz

Q7-What is the primary purpose of a **while** loop in Python?

A. To define a function.

B. To create an infinite loop.

C. To iterate over a sequence of elements.

D. To repeatedly execute a block of code as long as a condition is true.

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